

that have different colours, and mix these together, these will mix randomly. Nothing tells the white marbles to move to one side and black marbles to move to the other. Similarly, nanoparticles do not have any driving force to separate themselves in two. The team proved that polymer chains attached to the surface of these nanoparticles can drive this separation and self-assembly process.

AR tools to help healthcare workers save lives

Researchers from Purdue University have developed a unique augmented reality (AR)-based approach that allows experienced surgeons and physicians around the world to help less-experienced doctors in war zones, natural disasters and in rural areas perform complicated procedures.



The Purdue technique connects healthcare professionals in remote areas with more experienced surgeons and physicians

The technique features a transparent headset screen display that allows the mentee to see the patient in front of him or her, along with getting real-time on-screen feedback from the mentor. The mentor works from a separate location using a video monitor to view the AR feed and provide instant feedback to the field surgeon.

The AR headset worn by the mentee in the field is designed to replace the current telestrator technology, which uses a separate video screen and freehand sketches to provide feedback.

The system uses computer vision algorithms to track and align virtual notes and marks from the mentor with the surgical region in front of the mentee.

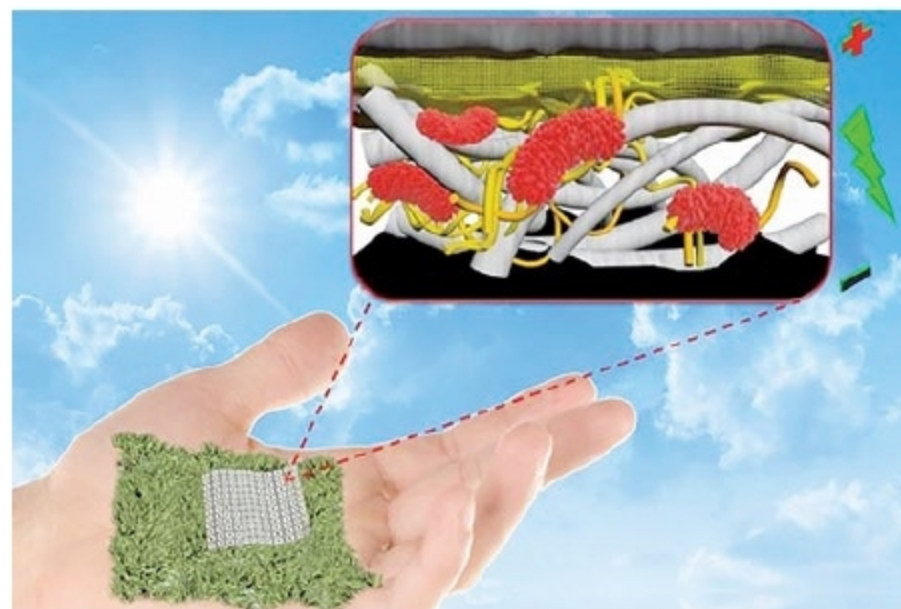
Scientists create biodegradable, paper-based biobatteries

Researchers at Binghamton University, State University at New York have created a biodegradable, paper-based battery that is more efficient than previously possible.

The biobattery uses a hybrid of paper and engineered polymers. The polymers—poly (amic) acid and poly (pyromel-

litic dianhydride-p-phenylenediamine)—give it biodegrading properties. The battery was tested for degradation in water, and it biodegraded without any special facilities, conditions or introduction of other microorganisms.

The polymer-paper structures are lightweight, low-cost and flexible. Due to the batteries being flexible, power enhancement can be potentially achieved by simply folding or stacking the hybrid, flexible paper-polymer devices.



The biobattery uses a hybrid of paper and engineered polymers (Credit: Seokheun 'Sean' Choi)

There has been a dramatic increase in electronic waste. These hybrid paper batteries exhibit a much higher power-to-cost ratio than all previously reported paper-based microbial batteries.

Associate professor Seokheun 'Sean' Choi from electrical and computer engineering department engineered the design of the paper-based battery, while professor Omowunmi Sadik from chemistry department made the battery a self-sustaining biobattery.

Smartphones may be used to better predict the weather

A recent study done by Tel Aviv University suggests that weather patterns that lead to flash floods may one day be tracked and anticipated by smartphones.

Prof. Colin Price of TAU's Porter School of the Environment and Earth Sciences says, "Sensors in smartphones are constantly monitoring the environment, including gravity, Earth's magnetic field, atmospheric pressure, light level, humidity, temperature, sound level and more. Vital atmospheric data exists today on some three to four billion smartphones worldwide. This data can improve the ability to accurately forecast the weather and other natural disasters."

To understand how smartphone sensors work, the team of researchers placed four smartphones around TAU's expansive campus under controlled conditions and analysed the data to detect phenomena such as atmospheric tides, which are similar to ocean tides. They also analysed data from an app called WeatherSignal.